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Experience

Research Intern – Database and Internet Privacy Lab at PSU - *May 2024 - September 2024*

- Engineered a workload generator in Java to synthesize 1,417+ queries and access control policies for an IoT environment spanning 5 user groups (36,000+ users), integrating it with a PostgreSQL database.
 - Evaluated the Sieve access control algorithm by indexing policies and creating guarded expressions, reducing enforcement overhead, enabling real-time, role-based access control for sensitive IoT data.
 - Presented work at PSU research showcase. Won 3rd place in Annual STEM Research Summit.
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Projects

AIFishbowl – Electron, MCP, STT/TTS, Voice Activation, GCP, Docker, NVIDIA Jetson Orin Nano, Python, JavaScript

- Deployed a conversational, multi-persona, agentic university assistant, hosted locally on dedicated hardware.
- Owned the entire application lifecycle, animations, real-time response streaming, and contributed to creating MCP tools. Containerized the application with Docker and hosted it on GCP.

Airline Reservation – Java, REST, Android Studio, Maven, HTML/CSS

- Built a full-stack airline booking system with a synchronized Android app and web frontend, implementing REST-based HTTP client-server communication with XML/JSON data exchange across all workflows.

Track and Sort – Electron/Nextron, Next.js, React, TypeScript, Tailwind, FastAPI, SQLite, PostgreSQL, argon2

- Track any lifestyle domain of yours. Designed a single schema-driven engine where an abstraction powers storage, adaptive forms, natural-language search, analytics, and AI capture. Developed a memory engine for frictionless data uploads.
- Built a bundled Python FastAPI sidecar (google-genai) that drafts structured entries from multimodal inputs.

Client-Server File Transfer – C, Socket Programming, TCP/IP, POSIX Threads, Mutexes, File I/O

- Engineered a file transfer system featuring concurrent file transfers, network IPC, asynchronous latency and delay simulation, thread-safe synchronization, IPv4 addressing, and a custom application layer.

Deep Learning – PyTorch, CUDA, MPS, NumPy, matplotlib, scikit-learn, Python – FCNN, CNN, RNN, VAE, ResNet50

- Trained, benchmarked architectures across image classification, generation, denoising, and time-series forecasting – 89.7% on FashionMNIST (top of a 24-config hyperparameter sweep), denoising to 5-60% pixel corruption, and a ResNet50 transfer study (10-1000 samples/class) – all GPU accelerated on CUDA and MPS.

LLM Sentiment Analysis – Python, HuggingFace Transformers, PEFT/LoRA, PyTorch, scikit-learn

- Fine-tuned Llama-3.2-1B for sentimental analysis of active social media data (X, Reddit), lifting macro-F1 from 0.58 (zero-shot) to 0.73 and benchmarking against zero-, one-, and 3-shot in-context learning.

Modular Retrieval-Augmented QA (RAG) – Python, DSPy, LangChain, Chroma, Gemini, Streamlit, Ollama, OpenAI

- Built the DSPy-based PDF RAG pipeline – LangChain chunking, Chroma vector store, top-k retrieval, chain-of-thought answers via Gemini 2.0 Flash, exposed with a Streamlit UI, spanning 3 frameworks of inputs (PDFs, BioASQ, Tech-corpora).
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Education

Portland State University - Portland, OR – B.S. in Computer Science | *Graduated March 2026*

- Operating Systems, Large Language Models, LLM Agents, Deep Learning, Data Clustering, Android and Java, Data with Python, Intro to Databases, Software Engineering Principles, Fundamentals of Software Engineering
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Skills

AI / LLMs [Python] – PyTorch, TensorFlow, HuggingFace Transformers, LoRA / PEFT, DSPy, LangChain, RAG, Ollama, STT/TTS

Machine Learning [Python] – CNNs, RNNs, MLPs, VAE, FVSBN, ResNet50, Computer Vision, CUDA & Metal Performance Shaders

Full-Stack [TypeScript | JavaScript | Java] – React, Next.js, Tailwind CSS, Framer Motion, Streamlit, Bootstrap, jQuery, Electron/Nextron, PixiJS, Android, FastAPI, Flask, Node.js, REST, Model Context Protocol (MCP).

Systems, Databases & Engineering [C | C++ | SQL | UNIX] – PostgreSQL, MySQL, SQLite, Chroma; FG-access control, Indexing

Operating Systems [C | C++] – Processes, Multithreading, fork, IPC/sockets, concurrency, ipv4 addressing, TCP/IP, CLI parsing

Tooling – Git/GitHub, GitHub Actions CI/CD, Docker, Maven, JUnit / pytest, TDD, conda / npm, pip, Vim.